

c) Availability of Credit

The ease with which credit is available in the economy is an important factor for economic growth. The eligibility and norms to avail credit from the financial system are set by the central bank or monetary authority of the country; hence, these norms are part of the monetary policy.

16.4 MONETARY POLICY INSTRUMENTS

The various instruments of monetary policy to control credit are divided in two categories, namely, Quantitative and Qualitative measures. Quantitative credit control measures are non-discriminatory in nature, say for example, when a certain interest rate is set by the central bank of a country, that rate applies to the banking system of the country as a whole. In contrast, Qualitative / Selective credit control measures vary from one section of the society to the other.

16.4.1 Quantitative Credit Control Measures

The important quantitative credit control instruments of the monetary policy are as follows:

- Bank Rate
- Open Market Operations
- Change in Minimum Reserve Ratio
- Change in Liquidity Ratio

a) Bank Rate

The most noticed and significant instrument of monetary policy is the rate of interest on borrowing and lending set by the central bank or monetary authority of the country. Bank rate is the rate of interest at which the central bank provides loans to commercial banks and other financial institutions. Increase in bank rate has the effect of increasing the rate of interest in the economy. Similarly, decrease in bank rate lowers the rate of interest in the economy. Higher bank rate lowers the extent of credit creation in the economy which leads to a decline in aggregate demand and hence lower prices. On the other hand, in a recessionary phase a lower bank rate is propounded. Borrowing from the central bank, usually for short period, by the commercial banks is done through the 'repo rate'. Similarly, commercial banks can deposit their excess liquidity in the central bank for which the 'reverse repo rate' is applicable.

b) Open Market Operations

Purchase and sale of securities in the open market by the central bank or the monetary authority is popularly known as open market operations. In order to contract the credit in the economy, the central bank sells securities in the open market. This leads to fall in aggregate demand and reduction in price level. Whereas, when credit is to be expanded, there is purchase of securities by the

central bank in the open market. This leads to increase in money supply which leads to increase in aggregate demand and production level in the economy.

c) Cash Reserve Ratio

The commercial banks are required to keep a certain percentage of their total deposits as cash with the central bank of the country. If the aim of the central bank is to reduce supply of money in the economy, it will raise the cash reserve ratio (CRR). An implication of higher CRR is that commercial banks are required to keep a higher percentage of their deposits in the form of cash with the central bank. Thus a lesser amount is left for lending purposes. In contrast, reducing the CRR would increase the credit creation in the economy as commercial banks have larger availability of funds with them for the purpose of lending and creating credit.

c) Statutory Liquidity Ratio

Statutory liquidity ratio (SLR) requires every bank to keep certain proportion of its total deposits in the form of liquid (or near liquid) assets. When the central bank needs to reduce credit, it raises the SLR. On the other hand, when the objective of the central bank is to expand credit, it reduces the SLR.

16.4.2 Qualitative Credit Control Measures

Very often the monetary authority in a country controls the supply of credit in the economy. Thus financial institutions may not extend credit to a borrower even if he/she has a demand for credit. These measures are selective in nature (because of it, they are often called selective credit control measures also) such that these measures target certain sectors of the economy or certain sections of the society. Secondly, it is discriminatory in the sense that different segments get differential treatments. Thirdly, there is a market failure in the sense that supply and demand for credit are not taken into consideration. Fourthly, there is disequilibrium in the credit market – supply and demand do not match.

We can classify qualitative nature can be classified in the following types:

- Change in margin requirement
- Credit rationing
- Direct action

a) Margin Requirement

Margin requirement is the margin between the value of goods pledged as security and the amount of loan. If the central bank finds that traders of any particular area are stock piling certain commodities which leads to an increase in prices of those commodities in the market, then the margin requirements of those

commodities are increased. This way credit is contracted and those defaulting traders are punished.

b) Credit Rationing

May a time the central bank resorts to credit rationing. Through rationing of credit, the central bank can perform the following tasks:

- It can decline loan to a particular commercial bank
- It can reduce the total amount of loan to all the commercial banks in general.
- It can fix a certain quota for different banks.

Hence, through the above selective credit rationing methods, the central bank can increase or reduce the credit in the market.

c) Direct Action

Central bank can take extreme actions against those commercial banks that disobey its directives. One such extreme action is directly imposing harsh restrictions on the credit of those commercial banks.

16.5 MONETARY POLICY IN INDIA

Reserve Bank of India (RBI) is the institution responsible for monetary policy framework in India. The policy targets and instruments of RBI have changed over the years in tandem with the domestic and global structural changes. Hence, in brief we discuss the evolution of monetary policy in India.

16.5.1 Early Years of Monetary Policy

Post Independence; say from 1950 to 1970, the RBI provided support to the government's plan expenditure using deficit financing. It was the industrialization phase when large scale public sector investments were financed by borrowing from the RBI. The RBI then focused on selective credit control check inflation while making sure that credit flowed to priority sectors. Since 1971, up to mid 1980s, the Indian government relied more on fiscal policy measures compared to monetary policy as Statutory Liquidity Ratio (SLR) and Cash Reserve Ratio (CRR) were increased. The high level of inflation in the 1980s led to the adoption of monetary targeting framework on the recommendations of the Sukhamoy Chakravarty Committee in 1985.

After the economic reforms of 1991, the RBI's role underwent a significant change. Steps were taken to curb the monetization of fiscal deficit. Say for example, by 1997 the ad-hoc treasury bills were completely phased out. However, monetization of fiscal deficit continued through primary issue of public debt in auctions. Since April 2006, the RBI is not allowed to subscribe to the primary issue of public debt.

16.5.2 Liquidity Adjustment Facility

Multiple indicator approach was adopted by RBI in 1998 in which high frequency financial market rates and fortnightly/ monthly indicators were tracked along with the output data for monetary policy purposes. In April 1999, on the recommendations of the Narasimham Committee (second committee), Interim Liquidity Adjustment Facility (ILAF) was introduced. Repo and reverse repo rates emerged as the primary policy rates of the RBI and the significance of CRR and SLR as instruments of RBI to manage liquidity got reduced. However, the ILAF suffered from non-existence of a ceiling rate and no unique policy rate. Hence, since May 2011 the repo rate has become the only independently varying rate under the Revised Liquidity Adjustment Framework (RLAF). In September 2014 the RLAF was fine-tuned following the recommendations of the Urjit Patel Committee. In May 2016, the Monetary Policy Framework Agreement (MPFA) was signed between the Government of India (GOI) and the RBI to regulate inflation and bring it below 4 percent in financial year 2016-17. In case of failure on the part of the RBI to comply with the above targets, it is accountable to the GOI and suggest reasons and remedies to achieve the same in future by a specified date.

Since February 2019 the RBI has followed an accommodative monetary policy in order to provide stimulus to the ailing economy. The RBI, however, realized that the Scheduled Commercial Banks (SCBs) do not transmit the rate cuts to the general public. In view of the above, the RBI in August 2019 decided to formalize the linkage between repo rate and bank lending rates.

16.6 TRANSMISSION MECHANISM OF MONETARY POLICY

As mentioned in the beginning of the Unit, monetary policy affects several key economic variables in the economy. Monetary policy decision, particularly on money supply and interest rate, sets in motion a sequence of effects. Transmission mechanism means the process or the channels through which monetary policy affects economic variables.

It is difficult to make a complete list of variables which get affected by a monetary policy decision. Economists have attempted to make a comprehensive list of the channels. There are differences amongst economists, however, on the importance of various channels. We have tried to compile the various channels (See Table 18.1) based on the studies of Mishkin (1995), Taylor (1995) and Boivin et al. (2010). Second, the impact of monetary policy on all variables is not uniform – while certain variables get influenced by a particular monetary policy decision to a great extent, the impact is negligible on certain other variables. It depends on the policy decision and its target variables. Third, some of these effects are visible immediately while certain others are noticeable with a lag.

Fig. 18.1: Monetary Transmission Mechanism

CHANNEL	CAUSATION CHAIN
Interest Rate Channel (a) Nominal Variant (b) Real Variant	Money $\uparrow$ $\rightarrow$ Interest Rate $\downarrow$ $\rightarrow$ Investment $\uparrow$ $\rightarrow$ Aggregate Demand $\uparrow$ Money $\uparrow$ $\rightarrow$ Expected Price $\uparrow$ $\rightarrow$ Expected Inflation $\uparrow$ $\rightarrow$ Real Interest Rate $\downarrow$ $\rightarrow$ Investment $\uparrow$ $\rightarrow$ Aggregate Demand $\uparrow$
Open Economy Channel Exchange Rate Channel	Money $\uparrow$ $\rightarrow$ Interest Rate $\downarrow$ $\rightarrow$ Exchange Rate $\uparrow$ (Depreciates) $\rightarrow$ Exports $\uparrow$ $\rightarrow$ Net Exports $\uparrow$ $\rightarrow$ Aggregate Demand $\uparrow$
Credit Channel (a) Bank Lending (b) Balance Sheet (c) Cash Flow (d) Unanticipated Price Level (e) Liquidity Effects	Money $\uparrow$ $\rightarrow$ Deposits $\uparrow$ $\rightarrow$ Bank Loans $\uparrow$ $\rightarrow$ Investment $\uparrow$ $\rightarrow$ Aggregate Demand $\uparrow$ Money $\uparrow$ $\rightarrow$ Price of Stocks (Net Worth) $\uparrow$ $\rightarrow$ Adverse Selection $\downarrow$ $\rightarrow$ Moral Hazard $\downarrow$ $\rightarrow$ Lending $\uparrow$ $\rightarrow$ Investment $\uparrow$ $\rightarrow$ Aggregate Demand $\uparrow$ Money $\uparrow$ $\rightarrow$ Interest Rate $\downarrow$ $\rightarrow$ Cash Flow $\uparrow$ $\rightarrow$ Adverse Selection $\downarrow$ $\rightarrow$ Moral Hazard $\downarrow$ $\rightarrow$ Lending $\uparrow$ $\rightarrow$ Investment $\uparrow$ $\rightarrow$ Aggregate Demand $\uparrow$ Money $\uparrow$ $\rightarrow$ Unanticipated Price of Stocks $\uparrow$ $\rightarrow$ Adverse Selection $\downarrow$ $\rightarrow$ Moral Hazard $\downarrow$ $\rightarrow$ Lending $\uparrow$ $\rightarrow$ Investment $\uparrow$ $\rightarrow$ Aggregate Demand $\uparrow$ Money $\uparrow$ $\rightarrow$ Price of Stocks $\uparrow$ $\rightarrow$ Value of Financial Assets $\uparrow$ $\rightarrow$ Likelihood of Financial Distress $\downarrow$ $\rightarrow$ Consumer Durables and Housing Expenditure $\uparrow$ $\rightarrow$ Aggregate Demand $\uparrow$
Other Channels (a) Wealth Effect (b) Tobin's q (c) Expectations	Money $\uparrow$ $\rightarrow$ Price of Stocks $\uparrow$ $\rightarrow$ Wealth $\uparrow$ $\rightarrow$ Consumption $\uparrow$ $\rightarrow$ Aggregate Demand $\uparrow$ Money $\uparrow$ $\rightarrow$ Price of Stocks $\uparrow$ $\rightarrow$ Tobin's q $\uparrow$ $\rightarrow$ Investment $\uparrow$ $\rightarrow$ Aggregate Demand $\uparrow$ Money $\uparrow$ $\rightarrow$ Expected inflation $\uparrow$ $\rightarrow$ Consumption/Investment $\uparrow$ $\rightarrow$ Aggregate Demand $\uparrow$

We now elaborate on some of the above transmission channels.

a) Interest Rate Channel

Nominal Interest Rate: When there is an increase in money supply it leads to a decline in nominal interest rate which further raises the incentives for investment. An increase in investment has obvious impact of increase in aggregate demand.

Real Interest Rate: An increase in money supply increases the expected price level, that is, it raises the rate of inflation in the economy leading to fall in the real rate of interest. The fall in interest rate raises investment and as a consequence the aggregate demand. The increase in expected price also provides an incentive to firms to increase production.

We know from rational expectations hypothesis that the average of short term rates is the long term rate. Therefore, transmission from short term rates to long term rates is essential for this transmission channel to be effective. For a developing economy like India, the costs of funding does not decrease in short term as substantial segment of the deposits in banks are long term in nature. Developing countries generally have an underdeveloped bond markets, less competition in the banking sector, and existence of a large informal sector for credit. These factors make the transmission mechanism through this channel extremely weak.

b) Open Economy Channel

Exchange Rate: An increase in money supply leads to fall in interest rate. Fall in interest rate leads to outflow of funds, as investors find it more profitable to invest abroad. This leads to decrease in supply of foreign currency which leads to depreciation of the domestic currency. Depreciation of domestic currency leads to increase in net exports leading to an increase in aggregate demand.

After the economic reforms of the 1990s, the Indian economy gradually opened up to the external sector. Therefore, the importance of this channel in monetary transmission in India is on the rise.

c) Credit Channel

Bank lending: An increase in money supply leads to an increase in bank deposits which enhance bank lending. Increase in bank lending leads to further investments; leading in turn to increase in aggregate demand.

Banks may be reluctant to lend despite the growing cash reserve due to depressed economic conditions, and some borrowers should be completely bank dependent in order to make this channel of transmission really effective. In countries like India, where financial markets are not fully developed as yet, this could be a significant channel of transmission. **Balance Sheet:** An increase in money supply raises the prices of the stocks as people invest money in stocks. This reduces the

drawbacks of asymmetric information which boosts confidence of the lenders and borrowers on each other. There is increased lending and thus investments in the growing economy which ultimately raises the aggregate demand.

Cash Flow: An increase in money supply leads to fall in interest rate and increase in cash flow in the economy. This again reduces the drawbacks of asymmetric information and increased investment and aggregate demand follows.

Unanticipated Price level: An increase in money supply raises the otherwise unexpected rise in price level of stocks in the economy. The drawbacks of asymmetric information are reduced with further raises lending, investment and aggregate demand.

Liquidity Effects: An increase in money supply raises the prices of stocks and value of other financial assets. This reduces the risk of financial distress further leading to increased consumption of consumer durables and hence increased overall aggregate demand.

d) Other Channels

Wealth effect: An increased money supply raises the price of stock and thus raising the wealth of the people. This increased wealth raises consumption and aggregate demand.

Tobin's Q Effect: 'Q' represents the ratio of the market value of a firm's existing shares (share capital) to the replacement cost of the firm's physical assets (thus, replacement cost of the share capital). An increase in the money supply raises the price of the stocks and thus raises the 'Q' value leading to increased investment and aggregate demand.

Expectations: An increased money supply raises the expectations of increased inflation in the economy which enhances present consumption and investment leading to increased aggregate demand.

16.7 LET US SUM UP

Monetary policy is the macroeconomic policy implemented by the central bank of the country in order to regulate credit through its various instruments. The credit control measures are classified into quantitative and qualitative. The quantitative measures are interest rates, open market operations, reserve requirements and liquidity ratio conditions. The qualitative measures are changes in margin requirements, rationing of credit and direct action. The transmission of monetary policy takes through various channels. These channels include interest rate channel, open market channel, credit channel and other channels such as expectations, Tobin's q and wealth effect. Every channel shows that an increased

money supply through various channels ultimately increase the aggregate demand in the economy. In developed countries the dominant channel of transmission is the traditional interest rate channel, whereas in developing countries the exchange rate and bank lending channel play a significant role.

16.8 ANSWERS/ HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) By monetary policy we mean a deliberate change in the interest rates and money supply by the central bank or monetary authority of the country.
- 2) The goals and targets of monetary policy are different and ultimately move towards the overall objective of economic policy in general.

Check Your Progress 2

- 1) There are two types of credit control measures of monetary policy, namely, Quantitative and Qualitative measures. The Quantitative measures are non discriminatory in nature and apply to the entire country, while the qualitative measures are selective or discriminatory in nature.

Check Your Progress 3

- 1) There are various channels of transmission of monetary policy.